

ANOMALY AND DEFECT TRACKING LOG

CSI-100 Cell Sorting Instrument (IVD)

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Log owner	Atif Lodhi, Systems Engineering
Product	CSI-100 Cell Sorting Instrument, all configurations

Purpose

Controlled export of all anomalies and defects recorded during verification and validation of the CSI-100, with severity, safety impact, root cause, resolution, change reference and regression status. Supports the V&V Summary Report (VVR-CSI-001) and the software release (IEC 62304 5.8.2).

Document Control

Export	Date	Purpose	Exported by
01	2026-10-31	Monthly export: start of system verification	A. Lodhi
02 to 06	2026-11-30 to 2027-03-31	Monthly exports	A. Lodhi
07	2027-05-14	Export for V&V Summary Report VVR-CSI-001 and software release 3.0.0 decision	A. Lodhi

This document is a controlled export of the anomaly records held in the defect tracking system. The tracking system is the master record; each export is reviewed and approved by the Anomaly Review Board before release.

Approvals

Function	Role	Signature / Date
Systems Engineering	Log owner, Anomaly Review Board chair	
Software Engineering	Software problem resolution owner	
Firmware Engineering	Firmware and FPGA problem resolution owner	
V&V Engineering	Verification lead	
Quality Engineering	Risk management and release approval	

Contents

1. Purpose and Scope
2. Problem Resolution Process
3. Classification
4. Record Fields and IEC 62304 Mapping
5. Summary
6. Trend Analysis
7. Anomaly Register
8. Detailed Anomaly Records
9. Release Assessment: Open and Deferred Anomalies
10. Reference Documents

1. Purpose and Scope

This log records every anomaly and defect found during verification and validation of the CSI-100, with its severity, safety impact, root cause, resolution, change reference and regression test status. It provides the records required by IEC 62304 clause 5.6.8 (entering integration and system test anomalies into the problem resolution process), clause 9 (software problem resolution) and clause 5.8.2 and 5.8.3 (evaluation and documentation of known residual anomalies at release).

The log covers anomalies from software unit, integration and system testing (SVP-CSI-010), system verification (DVP protocols in VVP-CSI-001), compliance testing, cybersecurity testing and design validation (VAL studies). Hardware, labeling and process anomalies found during V&V are recorded here as well, so that the V&V Summary Report has one complete source; hardware corrections are implemented through the engineering change process (ECO) and, where they affect production, transferred through DTP-CSI-001.

Anomalies found by developers before code enters integration are handled in the development tools and are not part of this log, as permitted by the Software Development Plan (SDP-CSI-010).

2. Problem Resolution Process

Every anomaly moves through the states below. The Anomaly Review Board (ARB) meets weekly and on demand for severity 1 anomalies. Transitions are recorded with user, date and reason in the tracking system.

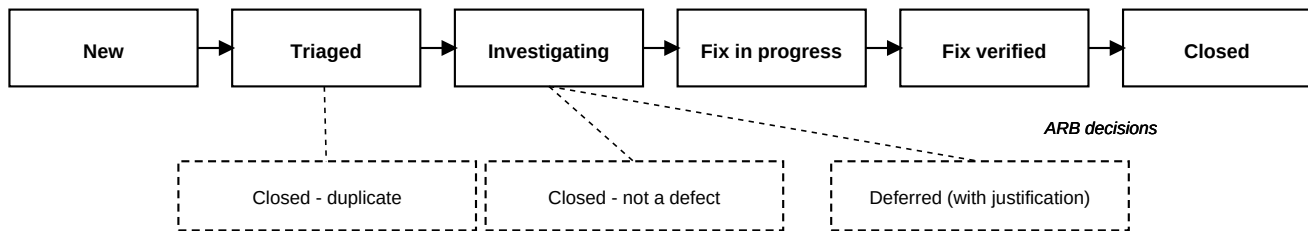


Figure 1. Anomaly states.

State	Entry condition	Responsible
New	Anomaly recorded by the person who found it, with evidence (test record, logs, screenshots)	Tester
Triaged	ARB assigns severity, safety impact, type, owner and target build	ARB
Investigating	Owner determines root cause and affected items, requirements and hazards	Owner
Fix in progress	Change request (CR) or engineering change order (ECO) approved	Owner, change board
Fix verified	Correction verified and regression tests passed on a controlled build	V&V
Closed	ARB confirms resolution, regression, risk file and document updates are complete	ARB, Quality
Closed - duplicate / not a defect	ARB confirms, with justification and reference	ARB
Deferred	Severity 3 or 4 only; ARB and Quality approve justification and risk evaluation; target release set	ARB, Quality

3. Classification

3.1 Severity

Severity follows VVP-CSI-001 Section 15.1 and is assigned by the ARB on the effect of the anomaly, not on the effort to fix it.

Severity	Definition	Release rule
1 Critical	Safety risk, or failure of a risk control	Must be closed before release; stops testing in the affected area
2 Major	Requirement not met	Must be closed before release
3 Minor	Requirement met, but unexpected or undesirable behavior	May be deferred with justification and risk evaluation
4 Cosmetic	No effect on function, performance or safety	May be deferred

3.2 Safety impact

Every anomaly is assessed against HA-CSI-001. If an anomaly relates to a hazard, the hazard is recorded, the risk management file is reviewed, and Quality must approve closure. All CSI-100 software items except SI-14 are Class B (SRS-SW-CSI-010 Section 2), so every software anomaly in a Class B item receives a root cause analysis and documented regression testing, whatever its severity.

3.3 Anomaly type and phase

Type	Examples
Software / Firmware / FPGA	Defects in workstation applications, controller firmware, FPGA logic
SOUP	Defects caused by third-party software; triggers SOUP anomaly evaluation update
Hardware / process / packaging	Design or manufacturing process defects found during V&V
Documentation / labeling	Labels, IFU, specifications
Test error	Anomaly caused by test setup, method or material; closed as not a defect with evidence
Validation finding	Anomaly found in a design validation study; traced to a user need

4. Record Fields and IEC 62304 Mapping

Field	Content	IEC 62304
ID, title, date found, found by, phase, test reference	Identification and source of the problem	9.1
Item, requirement, type	Affected software item or subsystem and requirement	9.1
Severity, safety impact (HA)	Criticality and relation to hazards	9.1, 9.2
Description, evidence	Behavior observed, conditions, attachments	9.1
Root cause	Result of investigation	9.2
Notified	Relevant parties advised, including risk management	9.3
Change reference (CR, ECO, DEV, DCR)	Link to change control	9.4
Status history	State changes with user, date, reason	9.5
Trend category	Used in Section 6	9.6
Fixed in, regression	Verification of the resolution and regression tests	9.7, 9.8

5. Summary

5.1 Status by severity

Status	1 Critical	2 Major	3 Minor	4 Cosmetic	Total
Closed	1	13	12	0	26
Closed - duplicate	0	1	0	0	1
Closed - not a defect	0	0	1	0	1
Open - fix verified	0	0	1	0	1
Open - investigating	0	0	1	0	1
Deferred	0	0	0	2	2
Total	1	14	15	2	32

5.2 Anomalies by area

Area	Total	Critical or major	Safety-related	Open or deferred
A. Research Application	8	3	3	1
B. Firmware	8	4	6	1
C. Admin Application	4	2	2	0
D. Data Management	3	1	0	1
Shared services	1	1	1	0
Hardware / packaging	2	2	2	0
Labeling	1	1	1	0
Validation finding	4	1	2	1
Test setup	1	0	0	0

5.3 Anomalies by phase found

Phase	Anomalies	IDs
Software integration	4	ANM-0003, ANM-0007, ANM-0034, ANM-0035
System verification	10	ANM-0011, ANM-0012, ANM-0014, ANM-0019, ANM-0032, ANM-0037, ANM-0039, ANM-0040, ANM-0047, ANM-0051
Software system test	10	ANM-0016, ANM-0023, ANM-0025, ANM-0026, ANM-0028, ANM-0042, ANM-0044, ANM-0045, ANM-0048, ANM-0053
Compliance testing	1	ANM-0021
Cybersecurity testing	1	ANM-0030
Design validation	6	ANM-0056, ANM-0058, ANM-0060, ANM-0062, ANM-0064, ANM-0066

Key figures: 32 anomalies recorded; 17 related to a hazard in HA-CSI-001; 1 critical and 14 major, all closed; 4 open or deferred, all severity 3 or 4 (Section 9).

6. Trend Analysis

IEC 62304 clause 9.6 requires analysis of problem reports to detect trends. Figure 2 shows anomalies opened and closed each month and the number open at month end.

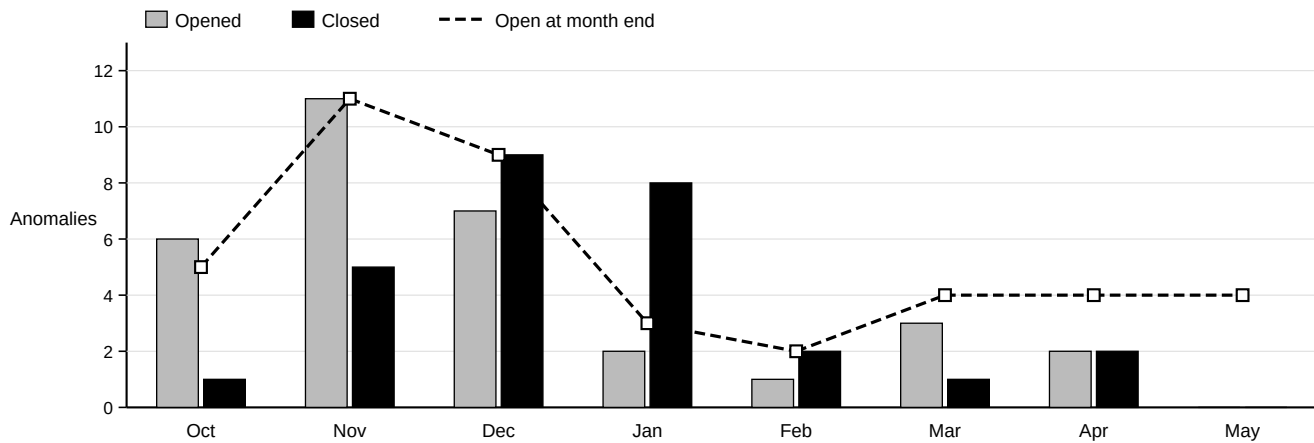


Figure 2. Anomalies opened and closed per month, October 2026 to May 2027 (May to export date).

Month	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Opened	6	11	7	2	1	3	2	0
Closed	1	5	9	8	2	1	2	0
Open at month end	5	11	9	3	2	4	4	4

6.1 Observations

- Most anomalies were found in the first three months of verification (October to December 2026), as expected when system verification and software system testing start together. The rate fell after RC4.
- Timing and sequencing in safety functions was a recurring theme (ANM-0003, ANM-0011, ANM-0023). Action: fault injection and timing tests were added to the nightly hardware-in-the-loop regression, and the fault reaction table in SAD-SW-CSI-011 Section 10.4 was reviewed.
- Configuration-dependent defects (ANM-0016, ANM-0026) came from code paths that bypassed the configuration rules. Action: all writes to configuration records now go through one service, and a test for every CFG-CSI-001 rule on every write path was added.
- Three hardware or process anomalies (ANM-0012, ANM-0021, ANM-0040) led to DMR and packaging changes that were transferred to manufacturing before the validation build, so validation units reflected the corrected design.
- Validation findings (ANM-0056 to ANM-0066) concerned use guidance, integration with site systems and long-run behavior, not requirement failures, which indicates that verification was effective.

7. Anomaly Register

One line per anomaly. Full details are in Section 8. Identifiers are assigned in sequence by the tracking system for the whole CSI-100 program; numbers not listed here were raised against development builds before integration and are outside the scope of this log (Section 1).

ID	Title	Item	Requirement	Sev.	Hazard	Found	Fixed in	Status
ANM-0003	Sort decision latency exceeds 20 μ s above 95,000 events/s	SI-11	FW-204, SYS-461	2	HA-07	2026-10-06	FPGA 3.0.2 (RC2)	Closed
ANM-0007	Safety supervisor removes HV enable during normal boot	SI-15	FW-604, FW-103	3	HA-02	2026-10-09	Safety MCU 3.0.1 (RC2)	Closed
ANM-0011	Clog response exceeds 2 s in 2 of 59 trials	SI-10	SYS-543, FW-602	1	HA-04	2026-10-21	Controller FW 3.0.3 (RC3)	Closed
ANM-0012	PI CV 3.1% on DV-02 on day 2	SS-03 (hardware)	SYS-401	2	HA-08	2026-10-22	WI-CSI-1002 Rev B	Closed
ANM-0014	Breakoff control oscillates with 130 μ m nozzle	SI-03	RA-501, SYS-613	3	—	2026-10-26	Research App 3.0.3 (RC3)	Closed
ANM-0016	Laser calibration factor applied to wrong laser on shared beam spot	SI-21	AD-202	2	HA-08	2026-10-28	Admin App 3.0.3 (RC3)	Closed
ANM-0019	Carryover 0.7% with 1 mL microtube holder	SI-04	SYS-471, RA-205	2	—	2026-11-02	Research App 3.0.3 (RC3)	Closed
ANM-0021	Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz	SS-06 (hardware), SI-11	SYS-503	2	HA-13	2026-11-04	Hardware rev B	Closed
ANM-0023	After link loss, Research App shows Ready while firmware is in safe state	SI-41, SI-01	FW-805, RA-1002	2	HA-02	2026-11-05	Research App 3.0.3, Admin App 3.0.3 (RC3)	Closed
ANM-0025	QC lockout window 25 h across daylight saving change	SI-05	RA-803, SYS-622	2	HA-08	2026-11-06	Research App 3.0.3 (RC3)	Closed
ANM-0026	480/20 filter rule not enforced after field upgrade to 445 nm	SI-20	AD-105	2	HA-08	2026-11-09	Admin App 3.0.3 (RC3)	Closed
ANM-0028	Audit trail missing old value for gate edits in draft assay template	SI-32	DM-201	2	—	2026-11-10	Records Service 3.0.3 (RC3)	Closed
ANM-0030	Replayed command accepted after reconnect (penetration test)	SI-10, SI-41	FW-802, FW-803	2	HA-08	2026-11-16	Controller FW 3.0.4, comms library 3.0.4 (RC4)	Closed
ANM-0032	Laser warning label missing on rear panel of 5-laser build	Labeling	SYS-513	2	HA-01	2026-11-18	LBL-CSI-001 Rev B	Closed
ANM-0034	PMT HV readback fault at 1,000 V	SI-12	FW-304	3	—	2026-11-20	Board FPGA 3.0.4 (RC4)	Closed
ANM-0035	Internal log loses last entries on power loss	SI-10	FW-703	3	—	2026-11-23	Controller FW 3.0.4 (RC4)	Closed

ID	Title	Item	Requirement	Sev.	Hazard	Found	Fixed in	Status
ANM-0037	384-well plate edge wells off target	SI-21, SS-08	SYS-483, AD-504	3	—	2026-11-30	Admin App 3.0.4 (RC4)	Closed
ANM-0039	Sample chamber takes 52 min to reach 4 °C at 30 °C ambient	SS-02 (hardware), SI-04	SYS-4101	3	—	2026-12-01	Controller FW 3.0.4 (RC4)	Closed
ANM-0040	DV-05 needed realignment after ISTA 3E transport test	SS-04 (hardware)	SYS-1201	2	HA-08	2026-12-07	PKG-CSI-001 Rev B	Closed
ANM-0042	Idle session lock not applied during calibration wizard	SI-21, SI-40	CM-103	3	—	2026-12-08	Admin App 3.0.4 (RC4)	Closed
ANM-0044	Memory growth in plotting library after 20 h acquisition	SI-01 (SOUP)	CM-303	3	—	2026-12-10	Research App 3.0.4 (RC4)	Closed
ANM-0045	FCS export missing \$PnV keyword for FSC	SI-30	DM-401	3	—	2026-12-11	Data Mgmt App 3.0.4 (RC4)	Closed
ANM-0047	Test failure: purity 96.8% on DV-01	Test setup	SYS-461	3	—	2026-12-14	—	Closed - not a defect
ANM-0048	Duplicate of ANM-0023	SI-01	RA-1002	2	HA-02	2026-12-15	See ANM-0023	Closed - duplicate
ANM-0051	Usability verification: clog recovery step skipped	SI-01	SYS-902, RA-504	3	HA-04	2027-01-14	Research App 3.0.5 (RC5)	Closed
ANM-0053	German UI text truncated in sort layout editor	SI-01	CM-305	4	—	2027-01-20	Planned 3.1	Deferred
ANM-0056	VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens	System	UN-02	2	HA-07	2027-02-24	Research App 3.0.6; OG-CSI-001 Rev C	Closed
ANM-0058	VAL-CSI-09 Site 2: LIS export rejected by site LIS	SI-30	UN-10, DM-404	3	—	2027-03-03	Data Mgmt App 3.0.6	Closed
ANM-0060	VAL-CSI-03: users unsure which nozzle is installed	SI-03, SS-03	UN-09	3	HA-08	2027-03-10	Research App 3.0.6; nozzle Rev C	Closed
ANM-0062	Precision study: Site 1 day 3 QC warning on violet array	SS-05, SI-05	UN-08	3	—	2027-03-17	—	Open - investigating
ANM-0064	Data Management search slow above 50,000 records	SI-30, SI-31	UN-10	4	—	2027-04-06	Planned 3.1	Deferred
ANM-0066	Plate voltage readback drift warning after 6 h sorting	SI-12, SS-07	FW-307	3	HA-02	2027-04-15	Board FPGA 3.0.6	Open - fix verified

8. Detailed Anomaly Records

ANM-0003	Sort decision latency exceeds 20 μs above 95,000 events/s
Found / phase	2026-10-06; Software integration; ITS-CSI-010 §11
Item / requirement	SI-11; FW-204, SYS-461
Severity / hazard / type	2 Major; HA-07; Firmware (FPGA)
Description	With 12 parameters and more than 95,000 events/s, 0.4% of sort decisions completed 22 to 26 μ s after the event window, so the charge was scheduled one drop late.
Root cause	Back-pressure from the event packetizer stalled the mask resolver stage when the DMA FIFO was more than 90% full. The sort path and data path shared a pipeline stage.
Resolution	Separated the sort decision path from the packetizer with an independent FIFO; the data path now drops to a lossless overflow buffer instead of stalling the sort path. Timing closure re-run.
Change reference	CR-SW-0412
Fixed in	FPGA 3.0.2 (RC2)
Regression and verification	UTS-CSI-010-11 timing suite; ITS §11 at 110,000 events/s: 0 late decisions in 10 ⁹ events; DVP-CSI-405 purity rerun pass.
Parties notified	Firmware, Systems, Quality (risk review)
Status	Closed, closed 2026-11-03

ANM-0007	Safety supervisor removes HV enable during normal boot
Found / phase	2026-10-09; Software integration; ITS-CSI-010 §10
Item / requirement	SI-15; FW-604, FW-103
Severity / hazard / type	3 Minor; HA-02; Firmware
Description	On every power-up SI-15 logged a heartbeat timeout because SI-11 does not send heartbeats while its FPGA image is loading. No unsafe condition: HV is off during boot.
Root cause	SI-15 started monitoring before SI-11 configuration completed. Requirement FW-103 (start in safe state) was met, but the false fault masked real boot faults in the log.
Resolution	SI-15 enters monitoring when SI-10 signals boot complete; during boot SI-15 holds HV and charge enables off unconditionally. Risk review confirmed no change to HA-02 controls.
Change reference	CR-SW-0415
Fixed in	Safety MCU 3.0.1 (RC2)
Regression and verification	UTS-CSI-010-15; ITS §10 fault injection suite (heartbeat loss at boot, run and sort): all pass.
Parties notified	Firmware, Quality
Status	Closed, closed 2026-10-30

ANM-0011	Clog response exceeds 2 s in 2 of 59 trials
Found / phase	2026-10-21; System verification; DVP-CSI-503
Item / requirement	SI-10; SYS-543, FW-602
Severity / hazard / type	1 Critical; HA-04; Firmware
Description	On DV-02, the four clog actions completed in 2.3 s and 2.4 s in 2 of 59 trials; all other trials 1.4 to 1.8 s. The requirement is a risk control for aerosol exposure.
Root cause	The fault manager executed actions sequentially and waited for the aspirator drawer to report closed before unloading the tube. Drawer close time varied with motor temperature.
Resolution	Actions are now issued in parallel (stream off and HV off first, then drawer and unload together) with a 1.5 s watchdog on completion. Motor acceleration profile corrected in SI-12.
Change reference	CR-SW-0421, CR-SW-0422
Fixed in	Controller FW 3.0.3 (RC3)
Regression and verification	DVP-CSI-503 repeated: 59 trials × 3 units, max 1.2 s, 0 failures; DVP-CSI-504 containment rerun pass; STS §B regression pass.
Parties notified	Firmware, Systems, Quality, Regulatory; RMR-CSI-001 updated
Status	Closed, closed 2026-12-02

ANM-0012	PI CV 3.1% on DV-02 on day 2
Found / phase	2026-10-22; System verification; DVP-CSI-401
Item / requirement	SS-03 (hardware); SYS-401
Severity / hazard / type	2 Major; HA-08; Hardware / process
Description	One of 27 runs gave a CV of 3.1% for PI-stained CEN at 488 nm. All other runs 2.2 to 2.8%.
Root cause	Microscopic void in the optical gel between cuvette and objective on DV-02, found by inspection. Gel dispense volume at the low end of the pilot work instruction range.
Resolution	Gel dispense volume and cure time changed in WI-CSI-1002; void inspection added. DV-02 flow cell re-coupled. Transferred to manufacturing as DMR change.
Change reference	ECO-0187
Fixed in	WI-CSI-1002 Rev B
Regression and verification	DVP-CSI-401 repeated on DV-02: 27 runs, max CV 2.6%, 95/95 upper limit 2.7%; gel process OQ rerun.
Parties notified	Manufacturing, Quality, Systems
Status	Closed, closed 2026-11-18

ANM-0014	Breakoff control oscillates with 130 µm nozzle
Found / phase	2026-10-26; System verification; DVP-CSI-405
Item / requirement	SI-03; RA-501, SYS-613
Severity / hazard / type	3 Minor; No hazard; Software
Description	With the 130 µm setup, amplitude oscillated ± 3 V and Gap alternated between 11 and 14 pixels for up to 90 s after breakoff control was turned on.
Root cause	Fine-step gain in the breakoff controller was tuned for 70 to 100 µm nozzles; at 12 kHz the image response lag is longer and the loop overshoots.
Resolution	Controller gains made part of each nozzle's default setup; 130 µm gain reduced by 50%.
Change reference	CR-SW-0426
Fixed in	Research App 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-03; DVP-CSI-405 breakoff stability per nozzle pass; STS §A6 pass.
Parties notified	Software, Systems
Status	Closed, closed 2026-12-04

ANM-0016	Laser calibration factor applied to wrong laser on shared beam spot
Found / phase	2026-10-28; Software system test; STS-CSI-010 §C2
Item / requirement	SI-21; AD-202
Severity / hazard / type	2 Major; HA-08; Software
Description	On a 6-laser unit, calibrating the 445 nm laser wrote its factor to the 488 nm laser record, because both share a beam spot.
Root cause	Calibration records were keyed by beam spot, not by laser serial number.
Resolution	Calibration records keyed by laser serial number; migration script re-keys existing records and flags any that cannot be matched for recalibration.
Change reference	CR-SW-0428
Fixed in	Admin App 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-21; STS §C2 all laser combinations of CFG-CSI-001; data migration test pass.
Parties notified	Software, Service, Quality
Status	Closed, closed 2026-11-25

ANM-0019	Carryover 0.7% with 1 mL microtube holder
Found / phase	2026-11-02; System verification; DVP-CSI-406
Item / requirement	SI-04; SYS-471, RA-205
Severity / hazard / type	2 Major; No hazard; Software
Description	With 1 mL microtubes, carryover was 0.6 to 0.7% (limit 0.5%). 12 × 75 mm and 15 mL tubes were 0.1 to 0.2%.
Root cause	Sample line outside flush time was fixed; the shorter sample probe immersion with microtubes leaves residue above the flushed length.
Resolution	Flush time and volume defined per tube holder type; holder type confirmed at Load.
Change reference	CR-SW-0431
Fixed in	Research App 3.0.3 (RC3)
Regression and verification	DVP-CSI-406 repeated all tube types, 20 sequences per unit: 95% upper bound 0.24%.
Parties notified	Software, Systems
Status	Closed, closed 2026-12-08

ANM-0021	Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz
Found / phase	2026-11-04; Compliance testing; DVP-CSI-501 / SVR-CSI-001
Item / requirement	SS-06 (hardware), SI-11; SYS-503
Severity / hazard / type	2 Major; HA-13; Hardware
Description	During IEC 61326-2-6 radiated immunity testing, event rate rose by up to 8% from noise events on the violet array while field was applied.
Root cause	Coupling into the unshielded ribbon cable between violet detector array and ADC board.
Resolution	Shielded cable and ferrite added; ADC input filter changed; no firmware change needed.
Change reference	ECO-0191
Fixed in	Hardware rev B
Regression and verification	SVR-CSI-001 immunity retest pass with performance monitoring; DVP-CSI-402 detection regression pass.
Parties notified	Electrical, Compliance, Regulatory
Status	Closed, closed 2026-12-15

ANM-0023	After link loss, Research App shows Ready while firmware is in safe state
Found / phase	2026-11-05; Software system test; STS-CSI-010 §B8
Item / requirement	SI-41, SI-01; FW-805, RA-1002
Severity / hazard / type	2 Major; HA-02; Software
Description	After a 5 s network disconnect during sorting, the application reconnected and showed Ready, while the instrument was latched in the safe state. The operator could not start a sort but was not told why.
Root cause	On reconnect, the communication library resubscribed to telemetry but did not request the current fault state; the latched fault message had been sent during the outage.
Resolution	Reconnect sequence now reads full instrument state (including latched faults) before enabling the user interface.
Change reference	CR-SW-0433
Fixed in	Research App 3.0.3, Admin App 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-41; ITS §9 reconnect suite; STS §B8 pass.
Parties notified	Software, Firmware, Human Factors
Status	Closed, closed 2026-12-01

ANM-0025	QC lockout window 25 h across daylight saving change
Found / phase	2026-11-06; Software system test; STS-CSI-010 §A8
Item / requirement	SI-05; RA-803, SYS-622
Severity / hazard / type	2 Major; HA-08; Software
Description	On the day clocks moved back, a performance check passed at 08:00 remained valid until 08:00 the next day local time, 25 h elapsed.
Root cause	Validity computed in local time instead of UTC.
Resolution	Validity computed in UTC as 24 h elapsed; display in local time.
Change reference	CR-SW-0434
Fixed in	Research App 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-05 time zone and DST cases; STS §A8 pass.
Parties notified	Software, Quality
Status	Closed, closed 2026-11-24

ANM-0026	480/20 filter rule not enforced after field upgrade to 445 nm
Found / phase	2026-11-09; Software system test; STS-CSI-010 §C1
Item / requirement	SI-20; AD-105
Severity / hazard / type	2 Major; HA-08; Software
Description	When a 445 nm laser was added as a field upgrade (AD-107), the configuration was saved with the 405 nm array still using 450/40.
Root cause	Rules engine ran on configuration edits but not on the upgrade path, which wrote laser records directly.
Resolution	Upgrade path now calls the rules engine before saving; saving is blocked until rules pass.
Change reference	CR-SW-0436
Fixed in	Admin App 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-20 all CFG-CSI-001 rules on edit and upgrade paths; STS §C1 pass.
Parties notified	Software, Service
Status	Closed, closed 2026-11-30

ANM-0028	Audit trail missing old value for gate edits in draft assay template
Found / phase	2026-11-10; Software system test; STS-CSI-010 §D2
Item / requirement	SI-32; DM-201
Severity / hazard / type	2 Major; No hazard; Software
Description	Editing a gate vertex in a draft IVD template created an audit entry with the new value but an empty old value.
Root cause	Gate objects were serialized after the edit was applied, so the previous state was lost before the audit event was built.
Resolution	Audit capture takes a snapshot before applying edits for all template object types.
Change reference	CR-SW-0437
Fixed in	Records Service 3.0.3 (RC3)
Regression and verification	UTS-CSI-010-32; STS §D2 audit completeness check across all record types pass.
Parties notified	Software, Quality (Part 11 assessment)
Status	Closed, closed 2026-12-03

ANM-0030	Replayed command accepted after reconnect (penetration test)
Found / phase	2026-11-16; Cybersecurity testing; DVP-CSI-701
Item / requirement	SI-10, SI-41; FW-802, FW-803
Severity / hazard / type	2 Major; HA-08; Firmware
Description	A captured valid 'set plate voltage' command replayed after a reconnect was accepted, because sequence numbers restarted at zero.
Root cause	Session keys were reused across reconnects and sequence numbers reset per connection.
Resolution	New session nonce and keys per connection; sequence numbers bound to the session.
Change reference	CR-SW-0440
Fixed in	Controller FW 3.0.4, comms library 3.0.4 (RC4)
Regression and verification	Penetration retest pass; UTS-CSI-010-10 and -41; ITS §9 pass.
Parties notified	Firmware, Software, Product Security
Status	Closed, closed 2027-01-12

ANM-0032	Laser warning label missing on rear panel of 5-laser build
Found / phase	2026-11-18; System verification; DVP-CSI-502
Item / requirement	Labeling; SYS-513
Severity / hazard / type	2 Major; HA-01; Documentation / labeling
Description	The rear service panel of the 5-laser build (DV-05) has Class 3B radiation behind it but no warning label.
Root cause	Label location drawing did not include the rear panel option used only on 5- and 6-laser builds.
Resolution	LBL-CSI-001 and label location drawing updated for all builds.
Change reference	ECO-0193
Fixed in	LBL-CSI-001 Rev B
Regression and verification	DVP-CSI-502 label inspection on all DV builds pass.
Parties notified	Regulatory, Manufacturing
Status	Closed, closed 2026-12-10

ANM-0034	PMT HV readback fault at 1,000 V
Found / phase	2026-11-20; Software integration; ITS-CSI-010 §10
Item / requirement	SI-12; FW-304
Severity / hazard / type	3 Minor; No hazard; Firmware (FPGA)
Description	Setting any PMT to 1,000 V raised a readback fault; readback was 1.3% low. Below 950 V readback was within 0.5%.
Root cause	Readback ADC scaling used a nominal divider value; at full scale the divider tolerance exceeds 1%.
Resolution	Per-board divider calibration stored in the board identity record and applied to readback.
Change reference	CR-SW-0442
Fixed in	Board FPGA 3.0.4 (RC4)
Regression and verification	UTS-CSI-010-12; ITS §10 full-range readback on all PMT boards pass.
Parties notified	Firmware, Manufacturing (calibration step added to FTS-CSI-001)
Status	Closed, closed 2027-01-08

ANM-0035	Internal log loses last entries on power loss
Found / phase	2026-11-23; Software integration; ITS-CSI-010 §10
Item / requirement	SI-10; FW-703
Severity / hazard / type	3 Minor; No hazard; Firmware
Description	In 3 of 20 power-cut tests, entries written 0.4 to 0.8 s before power loss were missing.
Root cause	Log flush was periodic (1 s) but the flash write took up to 300 ms; entries queued during the write were lost.
Resolution	Double-buffered log writes with power-fail interrupt flush.
Change reference	CR-SW-0443
Fixed in	Controller FW 3.0.4 (RC4)
Regression and verification	ITS §10 power-cut test: 100 cuts, no loss within 1 s window.
Parties notified	Firmware, Service
Status	Closed, closed 2027-01-10

ANM-0037	384-well plate edge wells off target
Found / phase	2026-11-30; System verification; DVP-CSI-407
Item / requirement	SI-21, SS-08; SYS-483, AD-504
Severity / hazard / type	3 Minor; No hazard; Software
Description	Single-bead deposition was 97.1% overall but 88% in columns 1 and 24 of 384-well plates.
Root cause	Plate stage calibration used first and last well only; stage has a small non-linearity near travel ends.
Resolution	Stage calibration adds a mid-plate point and applies a three-point correction.
Change reference	CR-SW-0446
Fixed in	Admin App 3.0.4 (RC4)
Regression and verification	DVP-CSI-407 repeated: 10 plates per format; 384-well 98.6%, edge columns 97%.
Parties notified	Software, Service
Status	Closed, closed 2027-01-19

ANM-0039	Sample chamber takes 52 min to reach 4 °C at 30 °C ambient
Found / phase	2026-12-01; System verification; DVP-CSI-408
Item / requirement	SS-02 (hardware), SI-04; SYS-4101
Severity / hazard / type	3 Minor; No hazard; Hardware
Description	Time to setpoint was 38 to 45 min at 15 to 25 °C ambient but 52 min at 30 °C (requirement 45 min).
Root cause	Peltier drive current limited to 70% by firmware default carried over from prototype.
Resolution	Drive limit raised to 90% after thermal analysis confirmed margin.
Change reference	CR-SW-0447, ECO-0195
Fixed in	Controller FW 3.0.4 (RC4)
Regression and verification	DVP-CSI-408 across 15 to 30 °C: max 41 min.
Parties notified	Mechanical, Firmware
Status	Closed, closed 2027-01-15

ANM-0040	DV-05 needed realignment after ISTA 3E transport test
Found / phase	2026-12-07; System verification; DVP-CSI-1201
Item / requirement	SS-04 (hardware); SYS-1201
Severity / hazard / type	2 Major; HA-08; Hardware / packaging
Description	After transport testing, the 561 nm spot on DV-05 had shifted; alignment bead rCV rose to 4.2% on that laser.
Root cause	Crate foam allowed resonance near 18 Hz on the launch plate axis.
Resolution	Crate foam redesigned and launch plate fastener torque increased; transport test repeated on two units.
Change reference	ECO-0197
Fixed in	PKG-CSI-001 Rev B
Regression and verification	DVP-CSI-1201 repeated on DV-03 and DV-05: installation qualification passed without realignment.
Parties notified	Packaging, Manufacturing, Service
Status	Closed, closed 2027-02-02

ANM-0042	Idle session lock not applied during calibration wizard
Found / phase	2026-12-08; Software system test; STS-CSI-010 §E1
Item / requirement	SI-21, SI-40; CM-103
Severity / hazard / type	3 Minor; No hazard; Software
Description	The Admin Application did not lock after the idle timeout while a calibration wizard dialog was open.
Root cause	Modal dialogs suppressed the idle timer events.
Resolution	Idle timer moved to the security client, independent of UI focus.
Change reference	CR-SW-0449
Fixed in	Admin App 3.0.4 (RC4)
Regression and verification	STS §E1 idle lock in all three applications and all dialogs pass.
Parties notified	Software, Product Security
Status	Closed, closed 2027-01-11

ANM-0044	Memory growth in plotting library after 20 h acquisition
Found / phase	2026-12-10; Software system test; STS-CSI-010 §E3
Item / requirement	SI-01 (SOUP); CM-303
Severity / hazard / type	3 Minor; No hazard; SOUP
Description	During the 24 h endurance test, memory grew by 1.6 GB and plot updates slowed to 1 per second after 20 h.
Root cause	Known anomaly in the plotting library version in use (texture cache not released on plot resize); not listed in the SOUP evaluation because it was fixed in a later patch release.
Resolution	SOUP updated to the patch release; SOUP register and anomaly evaluation updated.
Change reference	CR-SW-0450
Fixed in	Research App 3.0.4 (RC4)
Regression and verification	STS §E3 24 h endurance: memory stable, plots ≥ 2 updates/s; UI regression suite pass.
Parties notified	Software, Quality (SOUP register)
Status	Closed, closed 2027-01-20

ANM-0045	FCS export missing \$PnV keyword for FSC
Found / phase	2026-12-11; Software system test; STS-CSI-010 §D4
Item / requirement	SI-30; DM-401
Severity / hazard / type	3 Minor; No hazard; Software
Description	Exported FCS 3.1 files omitted the \$PnV (voltage) keyword for the FSC photodiode because it uses gain instead of voltage.
Root cause	Keyword writer skipped parameters without a PMT voltage.
Resolution	FSC gain written to \$PnV as required by FCS 3.1 guidance; custom keyword documents gain.
Change reference	CR-SW-0451
Fixed in	Data Mgmt App 3.0.4 (RC4)
Regression and verification	UTS-CSI-010-30; FCS validator on exports from all configurations pass.
Parties notified	Software
Status	Closed, closed 2027-01-09

ANM-0047	Test failure: purity 96.8% on DV-01
Found / phase	2026-12-14; System verification; DVP-CSI-405
Item / requirement	Test setup; SYS-461
Severity / hazard / type	3 Minor; No hazard; Test error
Description	One of 10 four-way sorts on DV-01 gave 96.8% purity.
Root cause	Investigation showed the bead mixture was prepared from an expired lot of one bead population with aggregates; sort re-analysis showed doublets from the sample, not sorting error.
Resolution	No product defect. Test repeated with a qualified lot under a documented deviation (DEV-VV-017). Bead lot check added to protocol.
Change reference	DEV-VV-017
Fixed in	—
Regression and verification	Sort repeated: 10 of 10 sorts > 98.9%; lower 95% bound 99.1%.
Parties notified	V&V, Quality
Status	Closed - not a defect, closed 2026-12-18

ANM-0048	Duplicate of ANM-0023
Found / phase	2026-12-15; Software system test; STS-CSI-010 §A10
Item / requirement	SI-01; RA-1002
Severity / hazard / type	2 Major; HA-02; Software
Description	Operator not alerted to latched fault after reconnect.
Root cause	Same cause as ANM-0023 (found on an RC2 build before RC3 was installed on that bench).
Resolution	Closed as duplicate; verified fixed on RC3.
Change reference	—
Fixed in	See ANM-0023
Regression and verification	See ANM-0023.
Parties notified	Software
Status	Closed - duplicate, closed 2026-12-16

ANM-0051	Usability verification: clog recovery step skipped
Found / phase	2027-01-14; System verification; DVP-CSI-901
Item / requirement	SI-01; SYS-902, RA-504
Severity / hazard / type	3 Minor; HA-04; Software (user interface)
Description	2 of 8 internal participants opened the sort block door before running aerosol evacuation in boost mode after a simulated clog.
Root cause	Recovery instruction appeared as a notification that could be dismissed; the boost step was the third item in a list.
Resolution	Clog recovery changed to a guided procedure that requires boost mode to run before the door-open step is shown; door-open warning added to the alarm text.
Change reference	CR-SW-0455
Fixed in	Research App 3.0.5 (RC5)
Regression and verification	DVP-CSI-901 repeated with 8 new participants: 0 skipped steps; STS §A5 pass. Use-related risk re-evaluated in HA-CSI-001.
Parties notified	Human Factors, Software, Quality
Status	Closed, closed 2027-02-05

ANM-0053	German UI text truncated in sort layout editor
Found / phase	2027-01-20; Software system test; STS-CSI-010 §E5
Item / requirement	SI-01; CM-305
Severity / hazard / type	4 Cosmetic; No hazard; Software (user interface)
Description	Three German labels in the sort layout editor are truncated at 1920 × 1080 resolution; full text shown in tooltips.
Root cause	Fixed-width label columns.
Resolution	Deferred to release 3.1 (layout rework). German release 3.0 labeling reviewed: no loss of meaning, tooltips give full text.
Change reference	CR-SW-0458 (planned 3.1)
Fixed in	Planned 3.1
Regression and verification	Not applicable for 3.0; tooltip check in STS §E5 pass.
Parties notified	Software, Regulatory (EU labeling)
Status	Deferred

ANM-0056	VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens
Found / phase	2027-02-24; Design validation; VAL-CSI-02
Item / requirement	System; UN-02
Severity / hazard / type	2 Major; HA-07; Validation finding
Description	Two apheresis specimens with CD34+ below 0.3% gave re-analysis purity of 93.1% and 93.8% (criterion 95%). All other specimens 96 to 99%.
Root cause	Specimens were run at 45,000 events/s to finish within the site's time window; at low target frequency, coincident events in Purity mode reduce purity. Instrument performed to specification; user guidance lacked a rate recommendation for rare populations.
Resolution	IFU updated with event rate guidance for rare-event sorting ($\leq 20,000$ events/s below 1% target frequency) and an in-software advisory when a sort population is below 1%. Two further specimens sorted per new guidance: 97.2%, 96.5%.
Change reference	CR-SW-0461, DCR-IFU-022
Fixed in	Research App 3.0.6; OG-CSI-001 Rev C
Regression and verification	VAL-CSI-02 supplementary run at Site 3 pass; STS §A7 advisory test pass. Not used as verification evidence.
Parties notified	Clinical Affairs, Human Factors, Software, Regulatory
Status	Closed, closed 2027-04-02
ANM-0058	VAL-CSI-09 Site 2: LIS export rejected by site LIS
Found / phase	2027-03-03; Design validation; VAL-CSI-09
Item / requirement	SI-30; UN-10, DM-404
Severity / hazard / type	3 Minor; No hazard; Validation finding
Description	Site 2 LIS rejected HL7 messages because the observation identifier used the local code without a coding system.
Root cause	HL7 export template allowed an empty coding system field; site LIS requires it.
Resolution	Coding system made a required field in the LIS export configuration with LOINC as default.
Change reference	CR-SW-0463
Fixed in	Data Mgmt App 3.0.6
Regression and verification	UTS-CSI-010-30; interface test with Site 2 LIS pass; VAL-CSI-09 re-run pass.
Parties notified	Software, Clinical Affairs
Status	Closed, closed 2027-03-24

ANM-0060	VAL-CSI-03: users unsure which nozzle is installed
Found / phase	2027-03-10; Design validation; VAL-CSI-03
Item / requirement	SI-03, SS-03; UN-09
Severity / hazard / type	3 Minor; HA-08; Validation finding
Description	4 of 30 participants hesitated at the nozzle confirmation prompt; none confirmed the wrong size, but two had to remove the nozzle to read its marking.
Root cause	Nozzle size marking is on the underside and not visible when installed.
Resolution	Marking added to the nozzle lever face (hardware change for next nozzle lot); prompt shows a picture of each nozzle. Residual use risk acceptable per HA-CSI-001.
Change reference	ECO-0203, CR-SW-0465
Fixed in	Research App 3.0.6; nozzle Rev C
Regression and verification	Formative check with 5 users pass; no new use errors. Human factors report updated.
Parties notified	Human Factors, Mechanical, Software
Status	Closed, closed 2027-04-20

ANM-0062	Precision study: Site 1 day 3 QC warning on violet array
Found / phase	2027-03-17; Design validation; VAL-CSI-04
Item / requirement	SS-05, SI-05; UN-08
Severity / hazard / type	3 Minor; No hazard; Validation finding
Description	Daily QC gave 'Pass with warning' for the violet 450/40 detector on one day; study results for that day remained within criteria.
Root cause	Under investigation. Preliminary: bead lot near expiry and a laboratory temperature excursion to 29 °C.
Resolution	Pending. Service data and bead lot retain testing in progress. No effect on validation acceptance criteria.
Change reference	—
Fixed in	—
Regression and verification	Pending.
Parties notified	Clinical Affairs, Service, Systems
Status	Open - investigating

ANM-0064	Data Management search slow above 50,000 records
Found / phase	2027-04-06; Design validation; VAL-CSI-09
Item / requirement	SI-30, SI-31; UN-10
Severity / hazard / type	4 Cosmetic; No hazard; Software
Description	At Site 2 (imported history of 60,000 records) free-text search took 8 to 12 s. Requirement has no search time limit; users rated it acceptable but slow.
Root cause	Free-text search not indexed.
Resolution	Deferred to 3.1: full-text index. Workaround: filter by date first (documented in release notes).
Change reference	CR-SW-0468 (planned 3.1)
Fixed in	Planned 3.1
Regression and verification	Not applicable for 3.0.
Parties notified	Software, Clinical Affairs
Status	Deferred

ANM-0066	Plate voltage readback drift warning after 6 h sorting
Found / phase	2027-04-15; Design validation; VAL-CSI-02
Item / requirement	SI-12, SS-07; FW-307
Severity / hazard / type	3 Minor; HA-02; Firmware / hardware
Description	At Site 3 a readback warning (1.6%, below the 2% fault limit) appeared after 6 h of continuous sorting; sorting unaffected.
Root cause	HV supply output drifts with temperature; warning threshold at 1.5% was tighter than needed.
Resolution	Fix implemented: temperature compensation of readback. Verified on bench; awaiting closure review.
Change reference	CR-SW-0470
Fixed in	Board FPGA 3.0.6
Regression and verification	UTS-CSI-010-12 and ITS §10 pass; 8 h sort soak on DV-02 pass. Closure review scheduled 2027-05-20.
Parties notified	Firmware, Electrical
Status	Open - fix verified

9. Release Assessment: Open and Deferred Anomalies

IEC 62304 clauses 5.8.2 and 5.8.3 require that known residual anomalies are evaluated to confirm they do not contribute to unacceptable risk, and that they are documented. The table lists every anomaly not closed at this export.

ID	Severity	Status	Risk evaluation	Disposition
ANM-0053	4 Cosmetic	Deferred	No hazard. Full text available in tooltips; meaning of labels preserved.	Deferred to release 3.1; listed in release notes
ANM-0062	3 Minor	Open - investigating	No hazard identified so far. QC warning did not affect results; QC lockout (RA-803) prevents IVD use after a QC failure.	Investigation to close before release decision; if cause is instrument-related, ARB re-evaluates
ANM-0064	4 Cosmetic	Deferred	No hazard. Search time does not affect results or safety.	Deferred to release 3.1; workaround in release notes
ANM-0066	3 Minor	Open - fix verified	HA-02 reviewed: warning is below the 2% fault limit; hardware interlocks and SI-15 remain the primary HV controls; no change in risk.	Fix verified; closure review 2027-05-20 before release
Release criterion (VVP-CSI-001, SDP-CSI-010)			Status at this export	
No open severity 1 or 2 anomalies			Met	
All safety-related anomalies closed or evaluated by Quality			Met: ANM-0066 evaluated, closure review scheduled	
All deferred anomalies justified and listed in release notes			Met: ANM-0053, ANM-0064	
Regression testing complete for all closed software anomalies			Met: see Section 8	
Risk management file updated for all anomalies related to hazards			Met: RMR-CSI-001 Rev A updated for ANM-0011, ANM-0051, ANM-0060	

Conclusion: subject to closure of ANM-0062 and ANM-0066 at the review on 2027-05-20, the anomaly status supports the release of software 3.0.0 and the V&V Summary Report. The final export (Export 08) will be issued with the release record.

10. Reference Documents

Document	Title
IEC 62304:2006 + A1:2015	Medical device software: software life cycle processes (5.6.8, 5.8.2, 5.8.3, clause 9)
ISO 14971:2019	Application of risk management to medical devices
VVP-CSI-001 Rev A	System Verification and Validation Plan
SRS-CSI-001 Rev D	System Requirements Specification
SRS-SW-CSI-010 Rev E	Software Requirements Specification
SAD-SW-CSI-011 Rev C	Software Architecture Description
SDP-CSI-010 Rev C / SVP-CSI-010 Rev B	Software Development Plan / Software Verification Plan
HA-CSI-001 Rev C / RMR-CSI-001	Hazard Analysis / Risk Management Report
DTP-CSI-001 Rev B	Design Transfer Plan
VVR-CSI-001	V&V Summary Report